

EE451 PHW4
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For the entire PHW, the test platform used was the USC CARC Discovery cluster. The programs were run using job scripts with configurations of `--nodes=1`, `--ntasks=4`, `--cpus-per-task=1`, `--mem-per-cpu=3GB`.

To run my code, after logging into HPC, first run:

`make`

to generate all the executables.

Then, run:

`sbatch job_p1.sl`

for problem 1.

Similarly, run:

`sbatch job_p2_1.sl`

`sbatch job_p2_2.sl`

`sbatch job_p2_3.sl`

for problem 2, approaches 1, 2, and 3.

The outputs from HPC can be found in `mpijob_p1.out`, `mpijob_p2_1.out`, `mpijob_p2_2.out`, `mpijob_p2_3.out`.

The output from HPC for problem 1 is shown below:

```
[rafaelwa@discovery2 PHW4]$ cat mpijob_p1.out
=====
SLURM_JOB_ID = 7027636
SLURM_JOB_NODELIST = d11-41
TMPDIR = /tmp/SLURM_7027636
=====
Process 0: Initially Msg = 451
Process 1: Msg = 452
Process 2: Msg = 453
Process 3: Msg = 454
Process 0: Received Msg = 454. Done!
```

The output from HPC for problem 2, approach 1 is shown below:

```
[[rafaelwa@discovery2 PHW4]$ cat mpijob_p2_1.out
=====
SLURM_JOB_ID = 7027705
SLURM_JOB_NODELIST = d05-26
TMPDIR = /tmp/SLURM_7027705
=====
Approach 1 - Process 0: Sum = 47126
```

The output from HPC for problem 2, approach 2 is shown below:

```
[[rafaelwa@discovery2 PHW4]$ cat mpijob_p2_2.out
=====
SLURM_JOB_ID = 7027706
SLURM_JOB_NODELIST = d05-26
TMPDIR = /tmp/SLURM_7027706
=====
Approach 2 - Process 0: Sum = 47126
```

The output from HPC for problem 2, approach 3 is shown below:

```
[[rafaelwa@discovery2 PHW4]$ cat mpijob_p2_3.out
=====
SLURM_JOB_ID = 7027707
SLURM_JOB_NODELIST = d05-26
TMPDIR = /tmp/SLURM_7027707
=====
Approach 3 - Process 0: Sum = 47126
```